

Isaiah Contreras

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EDUCATION

Texas State University

Bachelor of Science in Electrical Engineering
Ingram School of Engineering – ABET Accredited Program
Minor in Applied Mathematics
Minor in Business Management

San Marcos, TX
Expected Fall 2027

Central Texas College

Associate of Arts in Interdisciplinary Studies

Copperas Cove, TX
Graduated May 2024

ENGINEERING AND TECHNICAL EXPERIENCE

Amazon Transportation Operations Management Team (AUS2)

Pflugerville, TX

Transportation Area Manager Intern

[View TOM Optimization White Paper](#)

- Projected a network-wide recovery of 160,000 labor hours and \$6.4 million in annual cost avoidance by delivering 11 implementation-ready artifacts structured as a cost-effective full retention strategy.
- Directed daily yard logistics and 50+ Transportation Associates at an 820,000 sq. ft. fulfillment center, ensuring safety compliance and continuous freight flow across one of the largest yards in the Gulf Coast Region.
- Engineered a non-industry specific, 11-screen React/TypeScript UI prototype and developed the data-routing logic for a proposed real-time Slack integration, eliminating queue-based workflow latency to establish a closed-loop resolution system.
- Reduced Hostler Accept Delay by ~20% within an 8-week period, exceeding operational benchmark targets by executing Phase 1 and 2 operational quick wins, including visual yard signage and Vehicle Trailer Inspection (VTI) cards.

Semiconductor Science and Engineering Workshop at Texas State University

Texas State University - San Marcos, TX

Workshop Participant

- Gained hands-on experience in micro and nano device fabrication, analyzing Pulsed Laser Deposition (PLD) and sputtering techniques, and observing Hot Filament CVD (HFCVD) for diamond growth.
- Utilized state-of-the-art metrology tools—including SEM, AFM, and XRD—to evaluate semiconductor properties within cutting-edge research facilities funded by the NSF, DoD, and NASA.
- Earned a certification in Power, RF, and Digital testing from Keysight Technologies, and engaged with industry engineers from Intel and Samsung regarding advanced semiconductor manufacturing and enterprise technology transfer.
- Selected from a highly competitive applicant pool for a fully funded workshop, analyzing the integration of Artificial Intelligence, Machine Learning, and digital twins within modern semiconductor manufacturing.

GT World Challenge – SRO Motorsports America

Circuit of the Americas – Austin, TX

Scrutineering and Technical Operations Intern

- Extracted and processed vehicle telemetry from scrutineering dataloggers and on-board video systems to verify compliance and analyze on-track incidents during the 2025 GT World Challenge America race weekend.
- Utilized precision electronic tooling, including laser ride height instruments and handheld scanning systems, to conduct rapid tolerance checks and validate vehicle specifications against centralized database constraints.

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- Directed Parc Fermé operations for high-stakes track sessions, maintaining strict isolation protocols for post-session technical teardowns and fuel pump-out procedures to ensure systems-level compliance.

Tesla Cybertruck / Model Y Battery: Array and Pack
Production Associate

Gigafactory Texas - Austin, TX

- Contributed to Model Y production before transitioning to the initial launch of the Cybertruck, committing to 60-hour weeks to navigate massive operational pressure and complex logistics.
- Executed advanced troubleshooting and operational repairs for structural battery pack and array assemblies typically reserved for specialized technicians, rapidly resolving line bottlenecks across both vehicle platforms.
- Directed small, localized workstation teams as a certified workstation trainer, collaborating cross-functionally with engineering, technicians, quality, and production leadership to ensure strict production rate quotas were met during the Cybertruck launch window.

TECHNICAL PROJECTS

Engineering Sensor Data System | Python, Pandas, Matplotlib | <https://github.com/isaiah-con/Object-Oriented-EngineeringData-Analysis-System-NumPy-Pandas-5>

- Engineered a modular, object-oriented Python application to load, clean, and analyze large-scale datasets of simulated engineering sensor readings.
- Led the data integration and quality control phases, utilizing Pandas to filter, merge, and evaluate complex datasets against per-test metadata.
- Designed an interactive command-line interface (CLI) and implemented Matplotlib visualizations for statistical reporting, incorporating robust error handling to ensure data integrity.
- Authored comprehensive interactive test walkthroughs in Jupyter Notebooks to validate system functionality and structural software logic.

LEADERSHIP AND INVOLVEMENT

Texas State University IEEE Student Chapter

Sponsorship Coordinator

- Drive financial strategy and corporate outreach to expand the main chapter's general operating budget, securing vital capital to fund technical showcases, workshops, and student engineering projects.
- Pitch tiered sponsorship packages and corporate partnerships to local industry leaders, presenting the chapter's value proposition to secure financial backing and technology investments.
- Coordinate a multi-faceted resource strategy combining direct monetary funding, hardware and lab equipment donations, and exclusive corporate networking events to bridge the gap between students and industry.

Flight Logger Team Project (Bobcat Aerospace Subteam)

Firmware and Hardware Contributor

- Defined comprehensive design objectives, hardware tolerances, and software requirements for an advanced flight data logging system.
- Developed embedded systems firmware utilizing ESP-IDF in Visual Studio Code, configuring the operational environment and profiling hardware performance.
- Established Git-based version control workflows to streamline repository management and facilitate collaborative, cross functional materials research.

SKILLS

Software & Programming: Python (Pandas, NumPy, Matplotlib), C++, MATLAB, Jupyter Lab, ESP-IDF, SolidWorks, Git/GitHub, Visual Studio Code.

Hardware & Embedded Systems: Microcontroller firmware development, Microchip hardware integration, discrete digital logic (IC chips, logic gates, full adders, latches), Arduino, Soldering Station, HMI platforms.

Instrumentation & Testing: Keysight Semiconductor Industry-Ready Certified, Laser Ride Height Systems, Scanning Electron Microscopy (SEM), Atomic Force Microscopy (AFM), X-ray Diffraction (XRD), Multimeter, Calipers, Precision Measuring Tools.

Data Analytics & Operations: AWS QuickSight, Advanced Excel, Yard Management Systems (YMS), telemetry data acquisition, root cause data analysis.